

Approximative Solution Methods for Multiobjective Combinatorial Optimization

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Abstract

In this paper we present a review of approximative solution methods, that is, heuristics and metaheuristics designed for the solution of multiobjective combinatorial optimization problems (MOCO). First, we discuss questions related to approximation in this context, such as performance ratios, bounds, and quality measures. We give some examples of heuristics proposed for the solution of MOCO problems. The main part of the paper covers metaheuristics and more precisely non-evolutionary methods. The pioneering methods and their derivatives are described in a unified way. We provide an algorithmic presentation of each of the methods together with examples of applications, extensions, and a bibliographic note. Finally, we outline trends in this area.

Key Words: Multiobjective optimization, combinatorial optimization, heuristics, metaheuristics, approximation.

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1 Introduction

The last two decades have seen the development and the improvement of approximative solution methods – usually called “heuristics and metaheuristics” (Osman and Laporte (1996)). The success of metaheuristics, e.g., simulated annealing, tabu search, genetic algorithms, on hard single

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objective optimization problems is well recognized today. In an optimization context the term heuristic is used in contrast to methods that guarantee to find a global optimum such as, e.g., the “Hungarian Method” for solving the assignment problem or implicit enumeration schemes such as branch and bound and dynamic programming.

Although combinatorial optimization models have been successfully used in a vast number of applications, these models often neglect the fact that many real-life problems require taking into account several conflicting points of view corresponding to multiple objectives. Here are some examples:

- In portfolio optimization risk and return are the criteria that have generally been considered. Recently the classical Markowitz model has been criticized and other criteria have been mentioned: ratings by agencies, dividend, long-term performance, etc., see, e.g., Ehrgott et al. (2004).
- In airline operations, scheduling technical and cabin crew has a major effect on cost and small percentage improvements may translate to multi-million dollar savings. However, cost is not the only concern in airline operations. Robust solutions are desired, which avoid the propagation of delays due to crew changing aircraft, see Ehrgott and Ryan (2002).
- In railway transportation, the planning of railway network infrastructure capacity has the goals of maximizing the number of trains that can use the infrastructure element (e.g. a station) and to maximize robustness of the solution to disruptions in operation. This problem can be modelled as a large scale set packing problem with two objectives, Delorme et al. (2003).
- In radiation therapy planning for cancer treatment conflicting goals are to achieve a high dose level in the tumour whereas the dose absorbed by healthy tissue is to be limited. For anatomical and physical reasons these objectives cannot be achieved simultaneously. A multi-criteria model is described in Küfer et al. (2003).
- In computer networks, internet traffic routing may be enhanced if based on a multi-objective routing procedure to prevent network congestion. Multi-objective shortest paths between one router and all the

other routers of the network must be computed in real-time, by simultaneously optimizing linear objectives (cost, delay) and bottleneck ones (quality, bandwidth), see Randriamasy et al. (2002).

In combinatorial optimization the consideration of multiple objectives has received attention for about the last two decades but multiobjective combinatorial optimization (MOCO) has become a very active area of research only since the 1990's as shown in the bibliography by Ehrgott and Gandibleux (2000). Multiple objective (combinatorial) optimization differs from traditional single objective optimization in several ways:

- The usual meaning of the optimum makes no sense in the multiple objective case because a solution optimizing all objectives simultaneously does in general not exist. Instead, a search is launched for a feasible solution yielding the best compromise among objectives on a set of so-called efficient (Pareto optimal, non-dominated) solutions.
- The identification of a best compromise solution requires the preferences expressed by the decision maker to be taken into account.
- The multiple objectives encountered in real-life problems can often be expressed as mathematical functions of a variety of forms. I.e., not only do we deal with conflicting objectives, but with objectives of different structures.
- The multiple objectives add to the difficulty of combinatorial optimization problems so that MOCO problems are very hard to solve exactly (Ehrgott (2000)), even if they are derived from easy single objective optimization problems.

With increasing interest in multiobjective models and the difficulties encountered when solving multiobjective optimization problems, interest in approximative methods for solving multiobjective optimization problems arose. However, this interest is relatively recent. It began with the work of Schaffer (1984) on multiple objective genetic algorithms. The work of Serafini (1992) started a stream of research on multiple objective extensions of local search based metaheuristics. Since then research on these methods has mushroomed, giving birth to many ideas.

In the broad area of optimization, multiobjective approximation methods attract a growing community of researchers, witnessed by a growing number of publications. Several PhD and habilitation theses dealing specifically with these methods have been written, Coello (1996), Mira de Fonseca (1995), El-Sherbeny (2001), Godart (2001), Hansen (1998), Jaszekiewicz (2001c), Schaffer (1984), Ulungu (1993), and Zitzler (1999). Recently, international publications have dedicated special issues to these methods (Journal of Heuristics 6(3), 2000; Foundations of Computing and Decision Sciences 25(4), 2000 and 26(1), 2001; Lecture Notes in Economics and Mathematical Systems 535, 2004).

Numerous methods are inspired by Evolutionary Multiobjective Optimization (EMO). To give an idea about the strong activity in this field we remark that a repository dedicated to EMO, Coello (2004), counts more than 1600 entries. Lately four books dedicated to EMO have been published, Bagchi (1999), Coello et al. (2001), Deb (2001), Osyczka (2001), as well as the proceedings of the EMO international conferences, Fonseca et al. (2003), Zitzler et al. (2001). In addition, several surveys have been published regularly over the last 10 years, e.g. Coello (1999), Coello (2000), Fonseca and Fleming (1995), and Jones et al. (2002).

In this paper we present a review of approximation methods for solving multiobjective combinatorial optimization problems (MOCO), considering especially non-evolutionary methods. The paper is organised as follows. In Section 2 we provide the basic definitions of multiobjective (combinatorial) optimization and approximation methods. In Section 3 we briefly discuss issues related to the quality of approximations, such as bounds, performance ratios and other quality measures. In Section 4 we present the multiobjective versions of the greedy and local search heuristics and give some examples of their application. The main metaheuristic paradigms of evolutionary and neighbourhood search based are explained in Section 5. In Section 6 we briefly review multiobjective evolutionary methods. Sections 7 and 8 cover Simulated Annealing and Tabu Search Metaheuristics for MOCO. Section 9 is about a metaheuristic based on the ant colonies paradigm, that is becoming more and more popular for MOCO. In Section 10 we mention other techniques and the current trends in the field. Let us remark that we believe that our review covers all significant methods, although we may have missed some of the pertinent literature.

2 Basic Prerequisites

2.1 Multiobjective Optimization

A multiobjective optimization problem is defined as

$$\min_{x \in X} (f_1(x), \dots, f_p(x)), \quad (\text{MOP})$$

where $X \subset \mathbb{R}^n$ is a feasible set and $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is a vector valued objective function. By $Y = f(X) \subset \mathbb{R}^p$ we denote the image of the feasible set in the criterion space. We consider optimal solutions of (MOP) in the sense of efficiency (or Pareto optimality), that is, a feasible solution $x \in X$ is called efficient if there does not exist $x' \in X$ such that $f_k(x') \leq f_k(x)$ for all $k = 1, \dots, p$ and $f_j(x') < f_j(x)$ for some j . In other words, no solution is at least as good as x for all criteria, and strictly better for at least one.

Efficiency refers to solutions x in decision space. In terms of the criterion space, with objective vectors $f(x) \in \mathbb{R}^p$ we use the notion of non-dominance: If x is efficient then $f(x) = (f_1(x), \dots, f_p(x))$ is called non-dominated (or also efficient). The set of efficient solutions is X_E , the set of non-dominated vectors is Y_N . We may also refer to Y_N as the non-dominated frontier or the trade-off surface. For $y^1, y^2 \in \mathbb{R}^p$ we shall use the notation $y^1 \leq y^2$ if $y_k^1 \leq y_k^2$ for all $k = 1, \dots, p$; $y^1 \leq y^2$ if $y^1 \leq y^2$ and $y^1 \neq y^2$; and $y^1 < y^2$ if $y_k^1 < y_k^2$ for all $k = 1, \dots, p$. $\mathbb{R}_{\leq}^p$ denotes the nonnegative orthant $\{y \in \mathbb{R}^p : y \geq 0\}$, $\mathbb{R}_{>}^p$ is defined analogously.

To solve a multiobjective optimization problem means to find the set of efficient solutions or, in case of multiple x mapping to the same non-dominated point, for each $y \in Y_N$ find an $x \in X_E$ with $f(x) = y$. This concept of a set of efficient solutions is the major challenge of multicriteria optimization. Most methods require the repeated solution of single objective problems which are in some sense related to the multiobjective problem, see e.g. Miettinen (1999) or Ehrgott and Wiecek (2004).

In the following text we shall adopt the notation $x \succ x'$ if x dominates x' , i.e. if $f(x) \leq f(x')$.

2.2 Multiobjective Combinatorial Optimization Problems

Multiobjective combinatorial optimization problems can be formulated as follows:

$$\min \{Cx : Ax \geq b, x \in \mathbb{Z}^n\}. \quad (\text{MOCO})$$

Here C is a $p \times n$ objective function matrix, where c^k denotes the k -th row of C . A is an $m \times n$ matrix of constraint coefficients and $b \in \mathbb{R}^m$. Usually the entries of C , A and b are integers. The feasible set $X = \{Ax \geq b, x \in \mathbb{Z}^n\}$ may describe a combinatorial structure such as, e.g., spanning trees of a graph, paths, matchings etc. We shall assume that X is a finite set. By $Y = CX$ we denote the image of X under C in $\mathbb{R}^p$.

The biggest additional challenge in solving MOCOs as compared to multiobjective linear programmes (MOLPs) $\min\{Cx : Ax \geq b, x \geq 0\}$ results from the existence of efficient solutions which are *not* optimal for any scalarization using weighted sums

$$\min_{x \in X} \sum_{k=1}^p \lambda_k f_k(x), \quad (2.1)$$

called unsupported efficient solutions X_{NE} . Those that are optimal for some weighted sum problem (2.1) are called supported efficient solutions X_{SE} . A method called the 2-phase method has been applied to various problems to compute supported (in Phase 1) and unsupported (in Phase 2, using information obtained in Phase 1) efficient solutions, Ehrgott (1999), Lee and Pulat (1993), Ramos et al. (1998), Ulungu and Teghem (1994), Visée et al. (1998). Many methods generalizing single objective algorithms for the use with multiple objectives have been developed, see the survey on exact methods for MOCO in Ehrgott and Gandibleux (2000) and references in that bibliography.

An interesting observation in the context of heuristics is, that an exact algorithm which finds all supported solutions, i.e., solves (2.1) for all $\lambda \in \Lambda = \{\lambda \in \mathbb{R}_{>}^p : \sum_{k=1}^p \lambda_k = 1\}$, becomes a heuristic for determination of X_E . Also, a heuristic applied to solve (2.1) for some hard problem may actually yield a truly unsupported efficient solution.

From a methodological point of view unsupported efficient solutions are the main reason why the computation of X_E is hard, from a more theoretical point of view the computational complexity is another. Most (MOCO)

problems are *NP*-hard as well as $\#\mathcal{P}$ -hard as demonstrated by Ehrgott (2000). The best illustration of this fact is perhaps the unconstrained problem

$$\min_{x \in \{0,1\}^n} \left(\sum_{i=1}^n c_i^1 x_i, \sum_{i=1}^n c_i^2 x_i \right),$$

with $c_i^1, c_i^2 \geq 0; i = 1, \dots, n$ which is trivial with only one objective.

2.3 Approximation Methods

As in the single objective case, a reasonable alternative to exact methods for solving difficult MOPs is to derive an approximation method. An *approximation method* in a multiobjective optimization context is a method which finds either sets of locally potentially efficient solutions, that are later merged to form a set of potentially efficient solutions – the approximation – or globally potentially efficient solutions according to the current approximation. *Multiple objective heuristics (MOH)* and *multiple objective metaheuristic (MOMH)* are methods that aim to provide a good tradeoff between an approximation of the efficient solution set, denoted by *PE*, and the time and memory requirements to obtain them.

In Section 3 we discuss some theoretical issues related to approximation, such as bounds, performance ratios, and quality measures. In Section 4 we describe some genuine multiobjective heuristics. In this section we will not mention any methods to find the set X_{SE} . These are plentiful, but are all based on the principle of repeated application of single objective procedures (exact or heuristic) to (2.1) combined with a recursive dichotomic search procedure to generate the relevant values of λ . They are also very often restricted to two objectives.

Let us now define what we understand by heuristics and metaheuristics.

Heuristics. A heuristic is defined by Reeves (1995) as a technique which seeks good (i.e. near-optimal) solutions at a reasonable computational cost without being able to guarantee optimality, to state how close to optimality a particular feasible solution is or, in some cases, even to guarantee feasibility. Often heuristics are problem-specific, so that a method which works for one problem cannot be used to solve a different one.

Metaheuristics. In contrast, metaheuristics are powerful techniques applicable generally to a large number of problems. A metaheuristic refers to an iterative master strategy that guides and modifies the operations of subordinate heuristics by combining intelligently different concepts for exploring and exploiting the search space (Glover and Laguna (1997), and Osman and Laporte (1996)). A metaheuristic is a solution *concept*. The adaptation to a specific problem uses heuristics as solution methods.

A metaheuristic may manipulate a complete or incomplete single solution or a collection of solutions at each iteration. The family of metaheuristics includes, but is not limited to, constraint logic programming, genetic algorithms, evolutionary methods, neural networks, simulated annealing, tabu search, non-monotonic search strategies, greedy randomized adaptive search, ant colony systems, variable neighbourhood search, scatter search and their hybrids (Osman and Laporte (1996)). The success of these methods is due to their capacity “to solve in practice” some hard combinatorial problems.

3 Quality of Approximation

3.1 Bounds and Bound Sets

The quality of a solution of a combinatorial optimization problem can be estimated by comparing lower and upper bounds on the optimal objective function value. The success of optimization methods, in particular branch and bound, relies on the quality of available bounds.

In multiobjective optimization the concept of bounds is not well developed. The best possible lower and upper bounds on values of all non-dominated points are given by the ideal and nadir point y^I and y^N defined by

$$y_k^I = \min_{x \in X} f_k(x) = \min_{y \in Y_N} y_k, \quad k = 1, \dots, p$$

and

$$y_k^N = \max_{x \in X_E} y_k(x) = \max_{y \in Y_N} y_k, \quad k = 1, \dots, p,$$

respectively. We sometimes refer to a utopian point $y^U \leftarrow y^I - \varepsilon \mathbf{1}$, where $\mathbf{1}$ is a vector of all ones and ε is a small positive number. However, the

ideal and nadir points are usually far away from non-dominated points and do not provide a good estimate of the non-dominated set. In addition, the nadir point is hard to compute for problems with more than two objectives, see Ehrgott and Tenfelde-Podehl (2003).

To better capture the multiobjective nature of the problems and the fact that we are looking for a set of efficient solutions it is natural to generalize the notion of bounds to bound sets. The following definition is adapted from Ehrgott and Gandibleux (2001). For definitions of $\mathbb{R}_{\geq}^p$ -closedness and -boundedness see e.g. Sawaragi et al. (1985).

Definition 3.1. Let $\bar{Y} \subset Y_N$.

1. A lower bound set L for $\bar{Y}$ is an $\mathbb{R}_{\geq}^p$ -closed and $\mathbb{R}_{\geq}^p$ -bounded set $L \subset \mathbb{R}^p$ such that $\bar{Y} \subset L + \mathbb{R}_{\geq}^p$ and $L \subset (L + \mathbb{R}_{\geq}^p)_N$.
2. An upper bound set U for $\bar{Y}$ is an $\mathbb{R}_{\geq}^p$ -closed and $\mathbb{R}_{\geq}^p$ -bounded set $U \subset \mathbb{R}^p$ such that $\bar{Y} \in \text{cl} \left[(U + \mathbb{R}_{\geq}^p)^c \right]$ and $U \subset (U + \mathbb{R}_{\geq}^p)_N$.

Given a lower bound set and an upper bound set the quality of an approximation can then be estimated by the “distance” between the two sets. If we observe that a lower bound set is often a convex piecewise linear curve (obtained, e.g., by solving LP relaxations of (MOCO) and connecting the obtained points) and the upper bound set is given by a set of heuristically determined feasible solutions, the distance can be measured by the length of the orthogonal projection of the y values of the upper bound set on the linear pieces of the lower bound set. Alternatively, Tenfelde-Podehl (2002) proposes various measures of the area between a lower and an upper bound set for bicriteria problems.

Ehrgott and Gandibleux report first results on lower and upper bound sets in for the bicriteria knapsack problem (Ehrgott and Gandibleux (2001)) and for the bicriteria TSP, set covering and set packing problems, and the bicriteria assignment problem (Ehrgott and Gandibleux (2004)). Fernández and Puerto (2000) use bound sets in their exact and heuristic methods to solve the multiobjective uncapacitated facility location problem.

3.2 Performance Ratio

A lot of research has been carried out in the area of approximability of *NP*-hard combinatorial optimization problems, Ausiello et al. (1999). Surprisingly little is known about approximability of multicriteria problems. In single objective optimization, an approximation algorithm A is a (polynomial time) algorithm that finds a feasible solution x_A that is guaranteed to be within a certain ratio of the optimal solution x^* , i.e. $f(x_A) \leq r f(x^*)$. The constant $r \geq 1$ is often called the performance or approximation ratio.

In the multicriteria context, a new definition of performance ratio is needed. If we want to approximate the set X_E by one single heuristic solution, norms can be used. Ehrgott (2000) proposes ratios r_1 and r_2 if the solution x_A guarantees

$$\frac{\|f(x_A)\| - \|f(x^*)\|}{\|f(x^*)\|} \leq r_1,$$

respectively

$$\frac{\|f(x_A) - f(x^*)\|}{\|f(x^*)\|} \leq r_2$$

for all efficient solutions $x^* \in X_E$. An algorithm with performance ratio $r_1 = 1$ is obtained by solving

$$\min_{x \in X} \|f(x)\|,$$

which, however, can in itself be a hard problem. It is easy to see that in general algorithms with $r_1 < 1$ or $r_2 < 1$ are not possible.

A more interesting question of course is the approximation of X_E by sets PE with bounds on the difference of the set from the true efficient set. To achieve this, it is necessary to define a componentwise notion of approximation. A set of feasible solutions $X' \subset X$ is called ε -efficient (or ε -optimal) if for all $x \in X$ there is some $x' \in X'$ such that

$$f_k(x') \leq (1 + \varepsilon)f_k(x), \quad k = 1, \dots, p,$$

or $(f_k(x') - f_k(x)) \leq \varepsilon f_k(x)$, see Erlebach et al. (2002), Ruhe and Fruhwirth (1990), and Warburton (1987). This is the natural multiobjective version of the approximation ratio with the same ratio $r = (1 + \varepsilon)$ for all objectives.

An r -approximation algorithm A is an algorithm that runs in polynomial time and produces an $(r-1)$ -efficient solution set for a MOCO problem.

A polynomial time approximation scheme (PTAS) is a family of algorithms that contains for each $\varepsilon > 0$ a $(1 + \varepsilon)$ -approximation algorithm A_ε . If in addition A_ε is polynomial in ε^{-1} the family is called a fully polynomial time approximation scheme (FPTAS).

The existence of FPTAS for MOCO problems has been discussed by Safer and Orlin (Safer (1992), Safer and Orlin (1995a), Safer and Orlin (1995b)). They obtain results on the existence of FPTAS for some multicriteria network flow, knapsack, and scheduling problems. A general result has been given by Papadimitriou and Yannakakis (2000).

Ruhe and Fruhwirth (1990) (see also Burkard et al. (1989), and Fruhwirth et al. (1989)) present an algorithm which for given $r > 1$ finds an r -approximation of the efficient set of the (continuous) bicriteria network flow problem. Their method is based on an algorithm to approximate a convex curve (note that the efficient set of a bicriteria LP is a piecewise linear convex curve). The algorithm is therefore a pseudopolynomial approximation algorithm for the set X_{SE} of the integer bicriteria network flow problem.

Warburton (1997) and Hansen (1979) give FPTASs for the multicriteria shortest path problem. Erlebach et al. (2001) and Erlebach et al. (2002) develop a fast FPTAS for the multiobjective 1-dimensional knapsack problem and a PTAS for the multiobjective multi-dimensional knapsack problem (an FPTAS for the latter cannot exist unless $IP = NP$). In both papers, a partition of the objective space into intervals of increasing length is used. The fact that the knapsack algorithms are based on dynamic programming is another link between the two methods.

White (1986) proposes a number of definitions of ε -efficiency. Among them an additive version of the definition above, i.e. $\hat{x} \in X$ is called ε -efficient (or ε -optimal) if there is no $x' \in X$ such that $f(x') \leq f(\hat{x}) - \varepsilon \mathbf{1}$, where $\mathbf{1} \in \mathbb{R}^p$ is a vector of all ones. These definitions, however, have not found attention in MOCO literature.

3.3 Other Quality Measures

In the previous sections we have mentioned the distance between lower and upper bound sets and the performance or approximation ratios as quality measures for approximative solutions. For problems for which test

instances with known Pareto optimal solutions are available the quality of an approximation method can be estimated by the percentage of truly efficient points that it can detect.

There are a few other ideas in the literature. Kim et al. (2001) propose a new measure, the integrated convex preference (ICP), to compare the quality of algorithms for MOCO problems with two objectives.

Sayin (2000) proposes the criteria of coverage, uniformity, and cardinality as quality measures. Although developed for continuous problems the ideas may be interesting for MOCO problems. However, the methods proposed in Sayin (2000) can be efficiently implemented for linear problems only.

Other authors propose distance based measures, Viana and Sousa (2000) and visual comparisons of the generated approximations. The latter are restricted to bi-objective problems. Jaskiewicz (2001c) also distinguishes between cardinal and geometric quality measures. He gives further references and suggests preference-based evaluation of approximations of the non-dominated set using outperformance relations.

None of these measures have been universally adopted in the multiobjective optimization literature, and further research is clearly needed.

4 Heuristics for Multiobjective Combinatorial Optimization

Heuristics for MOCO problems can often be derived from heuristics for the single objective version of a combinatorial optimization problem. Two main strategies have been used. The first is to apply the single objective heuristic directly to solve (2.1) and combine it with a procedure to select appropriate values of λ .

We are interested in heuristics that deal with the multiple objectives directly. This can often be achieved by modifying the operators in the heuristic to deal with vectors rather than scalars. Thus, vector addition is used and the min or max operators are understood in the sense of the partial order $\leq$ and the preference relation $\succ$: $\min\{f(x) : x \in X\} = \{f(x) : x \in X \text{ such that there is no } x' \in X \text{ with } f(x') \leq f(x)\}$. Accordingly we write $\operatorname{argmin} \{f(x) : x \in X\} = \{x \in X : \text{there is no } x' \in X \text{ such that } x' \succ x\}$.

4.1 General Multiobjective Heuristics

In this section we describe two main heuristic principles in the multiobjective framework: the greedy and the local search principle. Let us consider $X \subset 2^A$ for some finite set A and assume, for ease of exposition that if $x \in X$ and $x' \subset x$ then $x' \in X$. Due to the partial order and the resulting incomparability of objective vectors, solutions are constructed in parallel (Algorithm 1).

Algorithm 1 Multiobjective Greedy Algorithm

MOGreedy: procedure (A, PE)

input : A , finite set

output : PE , the set of potentially efficient solutions

$PE \leftarrow \text{nondom}(A)$

while ($\exists x \in PE, a \in A \setminus x$ such that $x \cup a \in X$) **do**

for all $x \in PE, a \in \text{nondom}(A \setminus x)$ **loop**

if ($x \cup a \in X$) $PE \leftarrow PE \cup (x \cup a)$

end loop

$PE \leftarrow \text{nondom}(PE)$

end while

nondom(S): **procedure** (S)

$\text{nondom}(S) \leftarrow \{x \in S : \nexists x' \in S : f(x') \leq f(x)\}$

Such a constructive algorithm (without the filter for non-dominated vectors after the end of the for loop) has been used for the minimum spanning tree problem by Corley (1985). It performs reasonably well on problems whose single objective version can be solved optimally by the greedy algorithm. However, it does not necessarily generate all efficient solutions even for such problems. Serafini (1986) showed that for each efficient solution there is a topological order, such that if the greedy algorithm uses that topological order all efficient solutions of matroid problems are found. A greedy heuristic is also used by Rosenblatt and Sinuany-Stern (1989).

To describe the multiobjective local search heuristic we shall assume that a neighbourhood definition is given and that for any $x \in X$, $\mathcal{N}(x)$

defines the neighbourhood of x (Algorithm 2).

Algorithm 2 Multiobjective Local Search

MOLocalSearch: procedure $(x, \mathcal{N}, PE)$

input : x , feasible solution

output : PE , the set of locally potentially efficient solutions

$PE \leftarrow x, S \leftarrow x$

repeat

for all $\hat{x} \in S$ **loop**

for all $x \in \mathcal{N}(\hat{x})$ **loop**

if $(x \not\succeq \hat{x} \text{ and } \hat{x} \not\succeq x)$ $S \leftarrow S \cup x$

if $(x \succ \hat{x})$ $S \leftarrow S \cup x \setminus \hat{x}$

end loop

$S \leftarrow \text{nondom}(S)$

end loop

until $S = PE$

Andersen et al. (1996) give a local search heuristic based on the exchange of edges for the bicriteria spanning tree problem. Hamacher and Ruhe (1994) use a local search procedure in a two-phase algorithm for the bicriteria spanning tree problem after first calculating supported efficient solutions exactly.

The greedy and local search heuristics can be combined. The idea is to first construct solutions using the greedy approach and use local search to improve the solutions or generate more potentially efficient solutions. Such a combination is used by Sigal (1994) for the TSP. Both the multiobjective greedy algorithm and the multiobjective local search algorithm are of course general templates that can be applied to a wide variety of problems.

We shall now discuss heuristic methods for the multicriteria TSP as an example of how single objective heuristics can be adapted to the multicriteria case. Two of the best known heuristics for the TSP are the tree and Christofides' heuristic, Lawler et al. (1985). Both procedures begin by first constructing a minimal spanning tree. The tree heuristic then duplicates each edge of the spanning tree, thus obtaining a graph where all nodes have even degree. This graph is therefore Eulerian and has a tour that traverses

each edge exactly once. A Hamiltonian cycle can then be determined by elimination of nodes that are repeatedly visited.

Christofides' heuristic improves upon that by finding a minimal weight perfect matching of those nodes which have odd degree. The union of the spanning tree and the minimal weight matching is again Eulerian. From there the heuristic proceeds as the tree heuristic.

How can this be modified for the multiobjective TSP? Ehrgott (2000) replaces the steps of finding a minimal spanning tree and matching by finding a spanning tree and matching with minimal norm of the vector valued weights. He gives a number of results on the approximation properties of these algorithms according to the definitions of r_1 - and r_2 -performance ratios mentioned in Section 3.2.

5 Multiobjective Metaheuristics

The adaptation of metaheuristic techniques for the solution of multiobjective optimization problems has mushroomed over the last ten years, giving birth to multiobjective metaheuristic (MOMH). From a historical perspective, the pioneer approximation methods for multiobjective problems have appeared since 1984, in the following order: Genetic Algorithms (GA, Schaffer (1984)), Artificial Neural Networks (ANN, Malakooti et al. (1990)), Simulated Annealing (SA, Serafini (1992)), and Tabu Search (TS, Gandibleux et al. (1997)). The pioneer methods have two characteristics. First, they are inspired either by *Evolutionary Algorithms (EA)*, or by *Neighbourhood Search Algorithms (NSA)*. Second, the early methods are direct derivations of single objective optimization metaheuristics, incorporating small adaptations to integrate the concept of efficient solution for optimizing multiple objectives.

5.1 Evolutionary Algorithms versus Neighbourhood Search Algorithms

Evolutionary Algorithms manage a solution population $\mathcal{P}$ rather than a single feasible solution. In general, they start with an initial population and combine principles of self adaptation, i.e. independent evolution (as in the mutation strategy in genetic algorithms), and cooperation, i.e. the

exchange of information between individuals (as in the pheromones handled in ant colonies), to improve approximation quality. Because the whole population contributes to the evolutionary process, the generation mechanism is parallel along the frontier, and thus these methods are also called *global convergence-based methods*. This characteristic makes population-based methods very attractive for solving multiobjective problems.

In *Neighbourhood Search Algorithms*, generation relies upon one individual, a current solution x_n , and its neighbours $\{x\} \subseteq \mathcal{N}(x_n)$. Using a local aggregation mechanism for the objectives (often based on a weighted sum), a weight vector $\lambda \in \Lambda$, and an initial solution x_0 , the procedure iteratively projects the neighbours into the objective space in a search direction λ , by optimizing the corresponding parametric single objective problem. A local approximation of the non-dominated frontier is obtained using archives of the successive potentially efficient solutions detected. This generation mechanism is sequential along the frontier, producing a local convergence to the non-dominated frontier, and so such methods are called *local convergence-based methods*. The principle is repeated for diversified search directions to completely approximate the non-dominated frontier. NSAs present an aggressive convergence because the search is less dispersed, but they require more effort in diversification in order to cover the efficient frontier completely. All NSA methods use a current solution x_n and a neighbourhood $\mathcal{N}(x_n)$ to generate a new solution $x \in \mathcal{N}(x_n)$. The comparison of x and x_n according to p objectives $f_k(x)$, $k = 1 \dots p$ raises three possible situations (Figure 1 for a biobjective maximization problem) if $\Delta f_k = f_k(x) - f_k(x_n)$ is the difference between solutions x and x_n with respect to objective k .

(Case Ca) $\forall k \Delta f_k \geq 0$

All the objectives are improved for solution x . x (weakly) dominates the current solution x_n and is always accepted. If $\forall k \Delta f_k = 0$, the two solutions are equivalent. x is always accepted, except if only a *minimal complete set* X_{E_m} of efficient solutions (i.e. a subset of X_E that contains no equivalent solutions, and for any $x \in X_E$ there exists $x' \in X_{E_m}$ such that x and x' are equivalent) is computed.

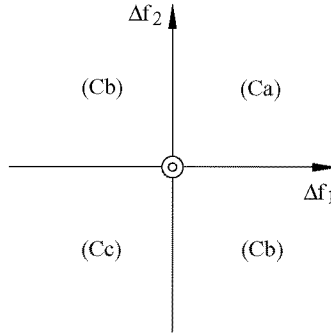


Figure 1: Three situations can occur when comparing solutions x and x_n according to multiple objectives (here, in the case of two maximized objectives).

(Case Cb) $\exists k, k' : \Delta f_k < 0$ and $\Delta f_{k'} > 0$

An improvement and a deterioration occur simultaneously for different criteria. Both solutions x and x_n are potentially efficient.

(Case Cc) $\forall k \Delta f_k \leq 0$

All objectives are deteriorated with at least one strict inequality. Solution x is dominated by x_n .

For the two last cases a scalarizing function $s(f(x), \lambda)$ is often used to project the multidimensional objective space into a monodimensional one using a weight vector $\lambda \in \mathbb{R}_{\geq}^p$. Such functions are well-known in the context of interactive procedures for multiobjective optimization. The scalarizing function allows to produce a “local aggregation” of the objectives in order to compute the “weighted distance” $\Delta s = s(f(x), \lambda) - s(f(x_n), \lambda)$ between $f(x)$ and $f(x_n)$.

Today, the multiobjective metaheuristics are often hybridized. For example, some NSA-based methods also handle a population of solutions, see Czyzak and Jaskiewicz (1996) and Hansen (1998). This coupling aims to break the independence aspect of the search process, inherent in the sequential generation principle, by exploiting information available in the population. One idea is to use the population as generator for a diversification strategy.

5.2 False Multiobjective Methods

Metaheuristics dealing with multiple objective optimization problems are sometimes wrongly presented as MOMH. This occurs when multiobjective problems are solved with a single objective strategy, looking for a unique compromise solution. In this case the original multiobjective problem is transformed or managed as the optimization of one or several single objective problems. The two following examples are representative of this class of methods. They are presented in Osman and Laporte (1996) as the first papers describing the use of TS as a technique for solving multiobjective problems.

A sequence of single objective problems. In Hertz et al. (1994) the method consists in solving a sequence of single objective problems considering in turn each objective f_j associated with a penalty term:

$$\begin{aligned} \min \quad & f_k(x) \\ \text{s.t.} \quad & f_j(x) \leq \bar{f}_j \quad j = 1 \dots k-1 \\ & x \in X. \end{aligned}$$

A parameterized value function. In Dahl et al. (1995) a family of scalarized problems (2.1) are solved to generate an approximation of a subset of X_E .

$$\begin{aligned} \min \quad & \lambda_1 f_1(x) + \lambda_2 f_2(x) + \lambda_3 f_3(x) \\ \text{s.t.} \quad & x \in X. \end{aligned}$$

We designate such approaches as “false multiobjective methods” and do not consider them in the framework of this synthesis.

6 The Multiobjective Evolutionary Algorithms Wave

The first to introduce a multiobjective metaheuristic was Schaffer (Schaffer (1984) and Schaffer (1985)). He developed a multiobjective evolutionary algorithm (MOEA), called *Vector Evaluated Genetic Algorithm (VEGA)*, which was an extension of Grefenstette’s GENESIS programme (Grefenstette (1984)) to include multiple objective functions. The vector extension concerns only the selection procedure.

6.1 Vector Evaluated Genetic Algorithm by Schaffer (1984)

For each generation in VEGA, three stages are performed (Algorithm 3). The selection procedure is performed independently for each objective. In the first stage, the population is divided into p subpopulations S^k according to their performance in objective k (routine `pickIndividuals`). Each subpopulation is entrusted with the optimization of a single objective. In the next stage, subpopulations are shuffled to create a mixed population (routine `shuffle`). In the final stage, genetic operators, such as mutation and crossover, are applied to produce new potentially efficient individuals (routine `evolution`). This process is repeated for N_{gen} iterations.

Algorithm 3 VEGA, Vector Evaluated Genetic Algorithm

```

input  :  pop,           the population size
          Ngen,         the limits of generations
          parameters,    the crossover probability and mutation rate
output :  PE,           the set of potentially efficient solutions

begin VEGA
--| Generate an initial population of pop individuals
   P0 ← initialization(pop)
--| Generation process
   for n in 1, ..., Ngen loop
     --| 1. Elaborate p sub-population of size pop/p using each objective k
     --| in turn
       Sk ← pickIndividuals( pop/p , k , Pn-1), ∀k = 1, ..., p
     --| 2. Set a population of size pop in shuffling together the p
     --| sub-populations Sk
       S ← shuffle(∪k=1,...,p Sk)
     --| 3. Apply genetic operators
       Pn ← evolution(S, parameters)
   endLoop
   PE ← PNgen
end VEGA
    
```

Because VEGA selects individuals who excel in one performance dimension without looking at the other dimensions, the speciation problem can arise with that method. This implies that individuals with a balanced performance on all objectives will not survive under this selection mechanism. Speciation is undesirable because it is opposed to the generation of compromise solutions. Due to this characteristic VEGA is labeled as a non-Pareto approach, Coello (1999). Additional heuristics were developed (like crossbreeding among the species) and studied to overcome this tendency.

6.2 Modern MOEAs

Although, it has some serious drawbacks, VEGA has had a strong influence up to now, and was at the origin of the Multiobjective Evolutionary Algorithm (MOEA) wave. Since VEGA many Multiobjective Evolutionary Algorithms have been developed. Significant progress concerns corrections of shortcomings observed in the first algorithms introduced and propositions of new algorithmic primitives to generate a better approximation of X_E . These MOEAs are characterized according to population structure, archiving, selection/elitism mechanism, and fitness function. Laumanns et al. (2001) give an overview of the techniques that are applied in most MOEAs (Figure 2).

Two central questions motivate the research about MOEA:

- (1) how to accomplish both fitness assignment and selection in order to guide the search toward the efficient frontier?
- (2) how to maintain a diversified population in order to avoid premature convergence and find a uniform distribution of solutions along the efficient frontier?

For the first question MOEAs are distinguished by the way the performance of individuals is evaluated in the selection. Firstly, when *the objectives are considered separately*, the selection of individuals is performed by considering each objective independently (Schaffer (1995)); or the selection is based on a comparison procedure according to a predefined (or random) order on the objectives (Fourman (1985)); or the selection takes into account probabilities assigned to each objective in order to determine a predominant objective (Kursawe (1992)). Secondly, when *the objectives are*

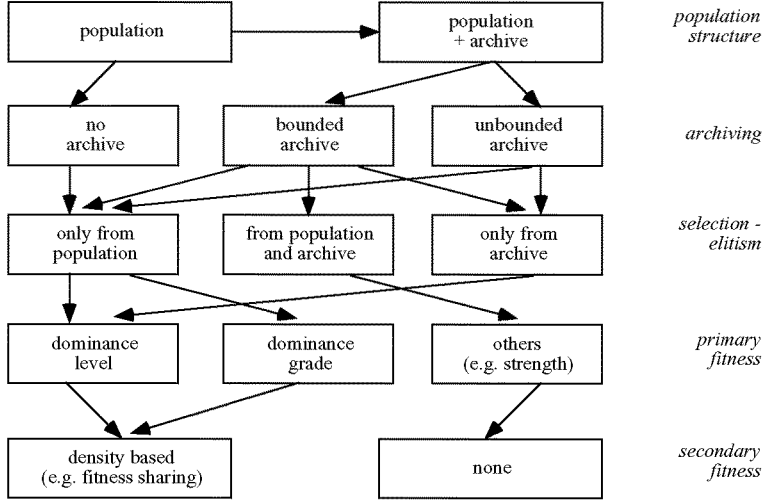


Figure 2: A taxonomy of techniques in MOEAs (Laumanns et al. (2001)). The arrows indicate existing combinations.

aggregated into a single parameterized objective function, the parameters of the function are systematically updated during the same runs (at random or using a particular weight combination) taking advantage of information collected on the population of individuals (Hajela and Lin (1992) and Murata and Ishibuchi (1995)). Each aggregation defines a search direction in the objective space and the idea is to optimize in multiple directions simultaneously. Thirdly, when *the concept of efficiency is directly used* (non-domination ranking) the fitness of an individual (i.e. a solution) is calculated on the basis of the dominance definition. The idea is to take advantage of information carried by the population of solutions using the notion of domination for selection. This is the most common approach and has led to several Pareto-based fitness assignment schemes, see Fonseca and Fleming (1993), Horn et al. (1994), Srinivas and Deb (1994), Zitzler and Thiele (1998), etc.

Goldberg (1989) has suggested the use of non-domination ranking according to the following principle (Figure 3). All non-dominated individuals are assigned rank 1. They are then temporarily withdrawn from the population. Then the remaining non-dominated individuals take rank 2 and so forth. The rank of an individual determines its fitness value. It is a concept of fitness related to the whole population. This evaluation scheme is im-

plemented in NSGA by Srinivas and Deb (1994). Other evaluations of the rank have been proposed. For Fonseca (Fonseca and Fleming (1993)) the rank of an individual is equal to the number of solutions which dominate this individual (implemented in MOGA93). For Zitzler and Thiele (1998) a strength value $s \in [0, 1]$ is calculated for each potentially efficient solution. For all other individuals in the population, the strengths of all potentially efficient solutions by which it is covered (dominated) are summed up.

The majority of the other components of a MOEA deal with the second question. A fitness sharing based on a principle of niches is the most frequently used technique. Most MOEAs are implementing fitness sharing, e.g., Fonseca and Fleming (1993), Horn et al. (1994), Srinivas and Deb (1994), Zitzler and Thiele (1998). *Niches* are solution neighbourhoods in objective space centered on candidate solutions and with a radius σ_{sh} . Based on the number N of solutions in these niches, the selection of individuals can be influenced to generate more in areas where niches are sparsely populated, with the goal of greater distribution uniformity along the non-dominated frontier (see Figure 4). A *sharing function*, which measures the distance $d(i, j)$ between a candidate solution i and a neighbour j , is defined by

$$\phi(d(i, j)) = 1 - \left(\frac{d(i, j)}{\sigma_{sh}} \right)^\alpha$$

if $d(i, j) < \sigma_{sh}$ and 0 otherwise. The parameter α amplifies ($\alpha > 1$) or attenuates ($\alpha < 1$) the sharing value computed. Thus the shared fitness of candidate i is

$$f_{s_i} = \frac{f_i}{\sum_{j=1}^N \phi(d(i, j))}$$

such that the fitness of candidates increases if ϕ values are small, i.e. the distance of neighbours from i is close to σ_{sh} .

Among the elements playing a significant role in a MOEA, the importance of the elite solutions to improve MOEAs is underlined by Laumanns et al. (2001). The results of numerical experiments show that the use of elitism must be accompanied by a strong rate of mutation in order to prevent the population from specializing too fast. The contribution of elite solutions in the generation of the non-dominated frontier in MOCO problems has been investigated by Gandibleux et al. (2001) and Morita et al. (2001). For the knapsack problem, using greedy solutions, or supported efficient solutions, in the population make the algorithm more apt to generate

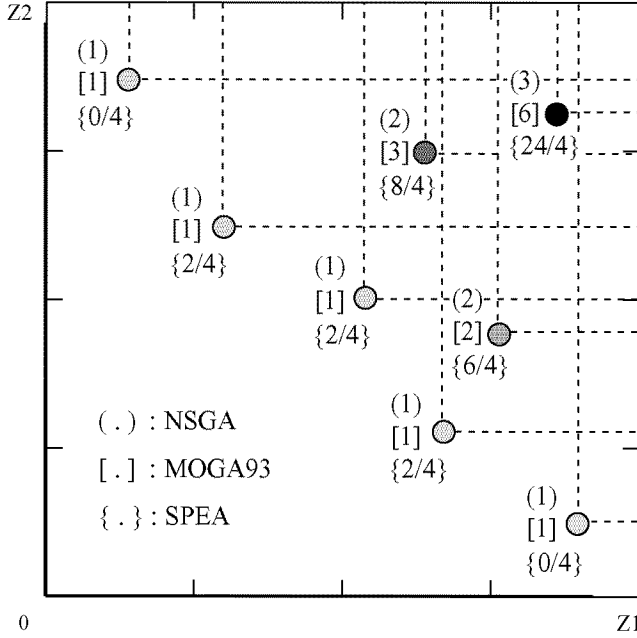


Figure 3: Computation of rank in NSGA (Srinivas and Deb (1994)), MOGA93 (Fonseca and Fleming (1993)) and SPEA (Zitzler and Thiele (1998)).

efficient solutions (Figure 5). Also, using elite solutions in combination with a generic path-relinking operator has shown impressive results in knapsack and assignment problems, Gandibleux et al. (2004).

6.3 Bibliographic Notes

A number of important implementations of MOEA have been published in recent years. There are even a number of surveys on the topic (see Coello (1999), Coello (2000), Fonseca and Fleming (1995), and Jones et al. (2002)). The most outstanding among the pioneer MOEAs are mentioned briefly below:

Vector Evaluated Genetic Algorithm (VEGA) by Schaffer (1984).

Multiple Objective Genetic Algorithm (MOGA93) by Fonseca and Fleming (1993). MOGA93 uses a ranking procedure in which the rank

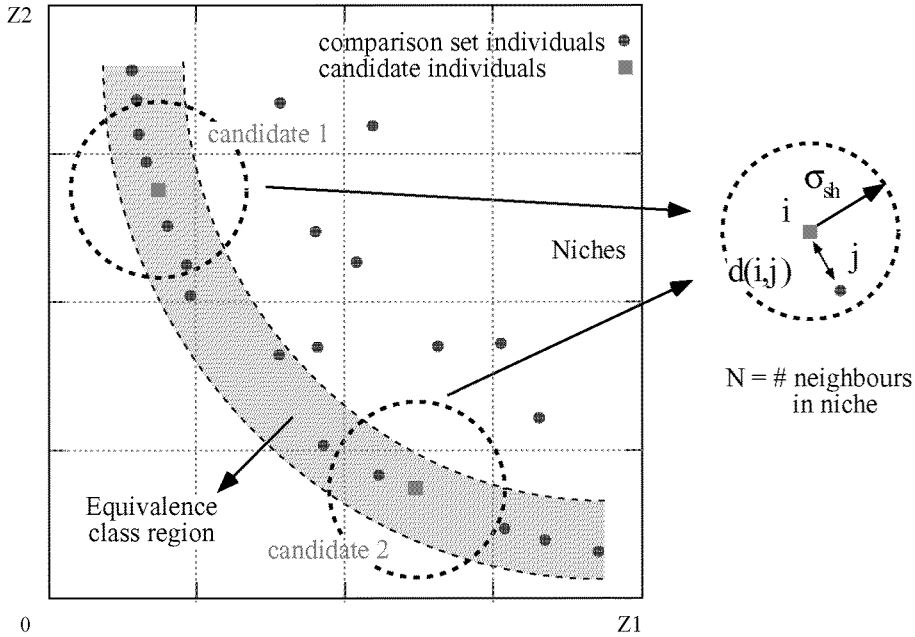


Figure 4: Niches are solution neighbourhoods in the objective space.

of an individual is equal to the number of solutions which dominate this individual.

Non-dominated Sorting Genetic Algorithm (NSGA) by Srinivas and Deb (1994). NSGA implements Goldberg's ranking idea in which the rank of an individual is equal to its domination layer, computed by ranking the population on the basis of domination.

Niched Pareto Genetic Algorithm (NPGA) by Horn, Nafpliotis and Goldberg, 1994 (Horn et al. (1994)). NPGA combines the Pareto dominance principle and a Pareto tournament selection where two competing individuals and a set of individuals are compared to determine the winner of the tournament.

For a long time, the problems investigated with these methods were often unconstrained bi-objective problems with continuous variables and non-linear functions. EA are appreciated by the engineering community which

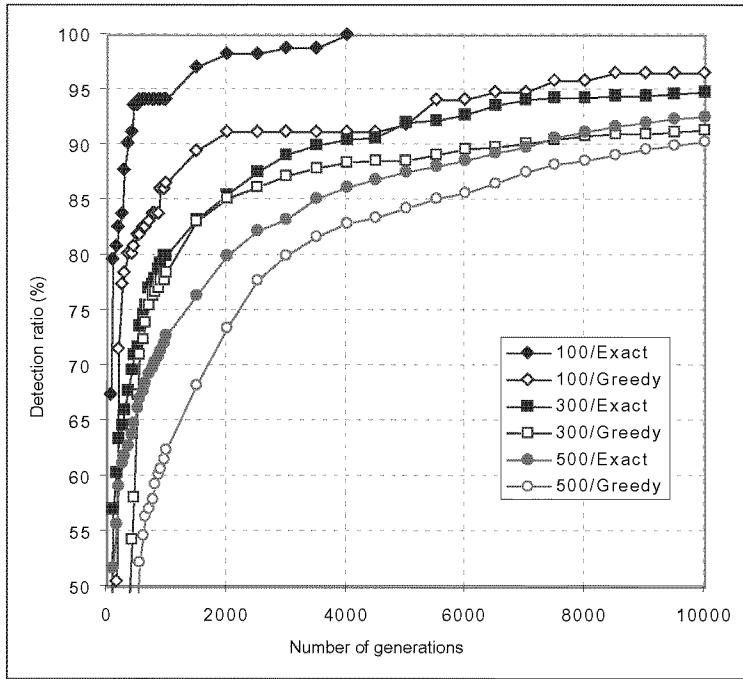


Figure 5: Influence of the quality of elite solutions (supported solutions and greedy solutions) on the detection ratio (number of exact solutions in the approximation set) in the case of bi-objective knapsack problems (100, 300 and 500 variables).

could explain the large number of MOEA applications for solving real world problems (in mechanical design or electronics, for example). Surprisingly, few MOEA have been applied to solve MOCO problems. A survey of the literature shows Gen and Li (1998a), Gen and Li (1998b) (Transportation Problems), Zhou and Gen (1999) (Spanning Tree Problems), Jaskiewicz (1998) (Travelling Salesperson Problems), Ben Abdelaziz et al. (1999), Gandibleux et al. (2001) (Knapsack Problems), Zitzler and Thiele (1999) (Multi-constraint Knapsack Problems), Liepins et al. (1990) (Set Covering Problems), Todd and Sen (1997) (Containership Loading Design), Morita et al. (2001), Pamuk and Köksalan (2001), Tamaki et al. (1994) (Scheduling Problems). The methods which have been used for MOCO are described below:

Multiple Objective Genetic Algorithm (MOGA) by Murata and Ishibuchi (1995). This method is not based on the Pareto ranking principle but on a weighted sum of objective functions, combining them into a scalar fitness function that uses randomly generated weight values in each iteration. Later, the authors coupled a local search with a genetic algorithm, introducing the memetic algorithm principle for multiobjective problems.

Method of Morita et al. (MGK) by Morita, Gandibleux and Katoh, 1998 (Gandibleux et al. (1998)). Seeding solutions, either greedy or supported efficient, are put in the initial population in order to initialise the algorithm with good genetic information. The biobjective knapsack problem is used to validate the principle. This method becomes a memetic algorithm when a local search has been performed on each new potentially efficient solution, Gandibleux et al. (2001).

Strength Pareto Evolutionary Algorithm (SPEA) by Zitzler and Thiele (1998). SPEA takes the best features of previous MOEAs and combines them to create a single algorithm. The multiobjective multi-constraint knapsack problem has been used as a benchmark to evaluate the method (Zitzler and Thiele (1999)).

Pareto Archived Evolution Strategy (PAES) by Knowles and Corne (1999). PAES is an evolutionary strategy that employs local search to generate new candidate solutions and a reference archive to compute solution quality.

Multiple Objective Genetic Local Search (MO-GLS) by Jaszkiewicz (2001b). This method hybridizes recombination operators with local improvement heuristics. A scalarizing function is drawn at random for selecting solutions, which are then recombined, and the offspring of the recombination are improved using heuristics.

Multiple Objective Genetic Tabu Search (MOGTS) by Barichard and Hao (2002). This is another hybrid method in which a genetic algorithm is coupled with a tabu search. MOGTS has been evaluated on the multi-constraint knapsack problem.

7 The Simulated Annealing Wave

Serafini (1992) was the first to use simulated annealing as a technique for multiobjective optimization problems. All multiobjective simulated annealing-based methods developed since then are still closely related to the original single objective method. They extend the single objective algorithm to cope with the notion of efficiency.

Often the authors tested various forms and definitions of acceptance rules. A lot of them have been suggested and discussed by Serafini (1992). Among those, four rules are frequently employed. If T_n is the current temperature value at iteration n the acceptance probability is:

- Rule C (Chebyshev rule):

$$p_n \leftarrow \min \left\{ 1 ; \exp \left(\max_{k=1 \dots p} \left\{ -\frac{\lambda_j (f_k(x) - f_k(x_n))}{T_n} \right\} \right) \right\} \quad (7.1)$$

- Rule SL (scalar linear rule):

$$p_n \leftarrow \min \left\{ 1 ; \exp \left(\sum_{k=1}^p \left\{ -\frac{\lambda_j (f_k(x) - f_k(x_n))}{T_n} \right\} \right) \right\} \quad (7.2)$$

- Rule W (weak rule):

$$p_n \leftarrow \min \left\{ 1 ; \exp \left(\min_{k=1 \dots p} \left\{ -\frac{\lambda_j (f_k(x) - f_k(x_n))}{T_n} \right\} \right) \right\} \quad (7.3)$$

- Rule P (product rule):

$$p_n \leftarrow \prod_{k=1 \dots p} \min \left\{ 1 ; \exp \left(-\frac{\lambda_j (f_k(x) - f_k(x_n))}{T_n} \right) \right\}. \quad (7.4)$$

Three methods are reported chronologically in this section, Czyzak and Jaskiewicz (1996), Parks and Suppapitnarm (1999), and Ulungu and Teghem (1992). It is important to emphasize that these methods have been elaborated independently from each other. The first method by Ulungu and Teghem (1992) uses a predefined set of weights $\lambda \in \mathbb{R}_+^p$. An independent SA process is then executed for each weight value. The second method by

Czyzak and Jaskiewicz (1996) introduces the use of a sample of solutions which are simultaneously optimized toward the efficient frontier while they are dispersed over the whole frontier. The last method by Parks and Suppapitnarm (1999) is not based on a principle of search directions, and it does not need an aggregation mechanism for the objectives. Each objective is considered separately. Advanced strategies using the population of potentially efficient solutions drive the approximation mechanism ensuring the detection of the whole efficient frontier.

7.1 Multiobjective Simulated Annealing by Ulungu (1992)

In 1992 (EURO XII conference, Helsinki), Ulungu introduced MOSA, one of the most popular simulated annealing methods for multiobjective optimization, see also Ulungu (1993). MOSA is a direct derivation of the simulated annealing principle for handling multiple objectives. Starting from an initial, randomly generated solution $x_{n=0}$ and a neighbourhood structure $\mathcal{N}(x_n)$, MOSA computes a neighbour $x \in \mathcal{N}(x_n)$ using a set of weights Λ that define search directions $\lambda \in \Lambda$. The comparison of x with x_n according to p objectives $f_k(x)$, $k = 1, \dots, p$ gives rise to three possible cases. If $\Delta f_k = f_k(x) - f_k(x_n)$ is the difference between solution x and x_n in the objective k (Figures 6 and 7 and Algorithm 4):

- (Ca) $\forall k \Delta f_k \leq 0$: All the objectives are improved for solution x . x_n (weakly) dominates x .
- (Cb) $\exists k, k' \Delta f_k < 0$ and $\Delta f_{k'} > 0$: Improvement and deterioration occur simultaneously for different criteria. Both solutions x and x_n are potentially efficient.
- (Cc) $\forall k \Delta f_k \geq 0$: All objectives are deteriorated with at least one strict inequality. Solution x is dominated by x_n .

A neighbour x is always accepted if it dominates x_n (Ca). When x is dominated (Cc), it can be accepted according to a decreasing probability, depending on the current “temperature” of the cooling schedule (Routine **isAccepted**). In the initial version of MOSA, a neighbour belonging to the intermediate situation (Cb) was also always accepted (Routine **isBetter** and Figure 6). This acceptance principle has been revised in a recent version of the method to include the search direction in the decision (Figure 7).

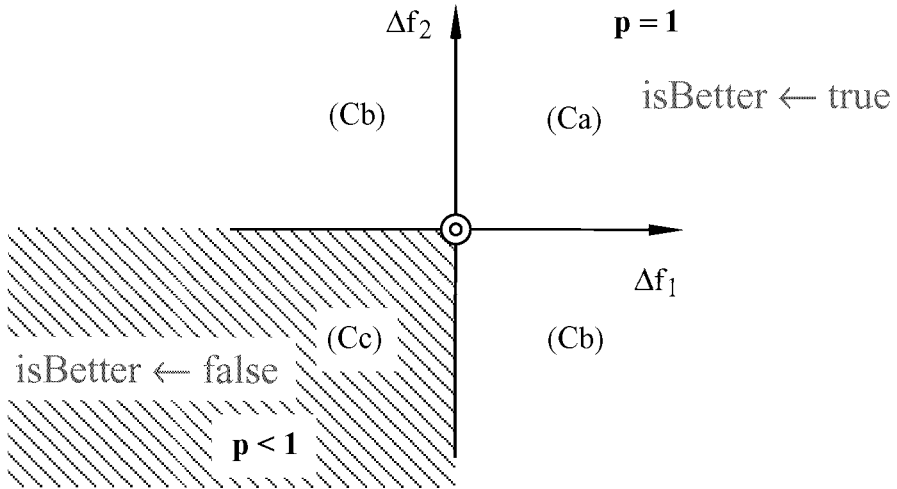


Figure 6: Acceptance principle in MOSA.

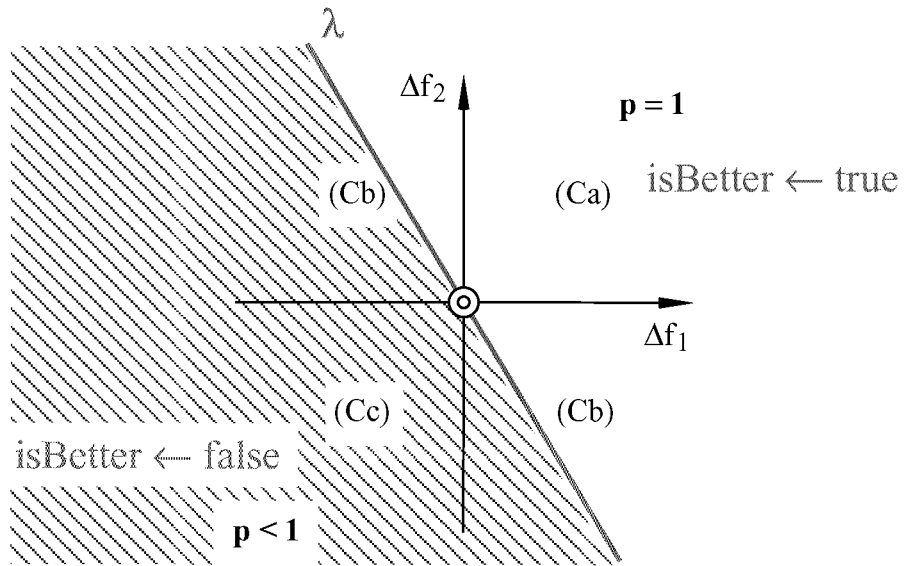


Figure 7: Revised acceptance principle in MOSA.

To measure the degradation in the routine `isAccepted`, the values are aggregated in a scalar using a scalarizing function $S(f(x), \lambda)$. Such a function makes a “local aggregation” of the objectives which allows the computation of the “weighted distance” $\Delta s = S(f(x), \lambda) - S(f(x_n), \lambda)$ between $f(x)$ and $f(x_n)$.

When a neighbour is accepted, the set of potentially efficient solutions PE_λ in direction λ is updated. The search stops after a certain number of iterations, or when a predetermined temperature is reached (Routine `isFinished`). At the end, MOSA combines the sequential processes in the objective space Y in a unique set PE by merging the sets PE_λ (Routine `merge`). The outline of MOSA for maximizing objectives is given in Algorithm 4.

The methods that follow are different primarily on four points: (1) the rule for acceptance of a new solution with some probability depending on the temperature; (2) the scheme for decreasing the temperature; (3) the mechanism that guides the browsing of the non-dominated frontier; (4) the use of information drawn from a population of individuals. Various forms and definitions of the acceptance rules have been tested by the method’s designers; many of these have been suggested and discussed in Serafini (1992). The most recent SA based methods include dynamic diversification mechanisms that exploit the set of potentially efficient solutions to drive the approximation process, Czyzak and Jaszekiewicz (1999).

7.2 Bibliographic Notes

The following methods based on simulated annealing have been published in the literature:

Multiobjective Simulated Annealing (MOSA) by Ulungu (1993).

The principle of MOSA is described in Ulungu’s PhD thesis, 1993. Numerical experiments are reported for three bi-objective combinatorial problems (assignment, transportation and knapsack problems). The method is presented later, Ulungu et al. (1995), in a systematic framework. Complete numerical experiments for the bi-objective knapsack problem are reported and compared with solutions obtained with an exact method. An interactive version of MOSA is derived in Teghem et al. (2000) and Ulungu et al. (1998). In Teghem et al.

Algorithm 4 MOSA, MultiObjective Simulated Annealing

input : Λ , set of weights
 $T, \alpha, N_{step}, T_{stop}, N_{stop}$, SA parameters
output : PE , set of potentially efficient solutions

begin MOSA

```

 $PE \leftarrow \emptyset$ 
for all  $\lambda \in \Lambda$  loop
   $T_0 \leftarrow T$  ;  $N_{count} \leftarrow 0$  ;  $n \leftarrow 0$ 
  randomly draw  $x_n \in X$  ;  $PE_\lambda \leftarrow \{x_n\}$ 
  repeat randomly draw  $x \in \mathcal{N}(x_n)$ 
    if isBetter( $x, x_n$ ) or else isAccepted( $x, x_n, n, T_n, \lambda$ ) then
       $PE_\lambda \leftarrow \text{archive}(PE_\lambda, x)$ ;  $x_{n+1} \leftarrow x$  ;  $N_{count} \leftarrow 0$ 
    else
       $x_{n+1} \leftarrow x_n$ ;  $N_{count}++$ 
    endIf
     $n++$  ; updateParameters( $\alpha, n, T_n$ )
  until isFinished( $N_{count}, T_n$ )
endLoop
 $PE \leftarrow \text{merge}(PE_\lambda)$ 

```

end MOSA

With :

- **isBetter**, a predicate that computes $\Delta f_k \leftarrow f_k(x) - f_k(x_n)$ and then returns $\neg(\forall k : \Delta f_k \leq 0)$.
- **isAccepted**, a predicate that computes the acceptance probability p_n for current iteration as $\exp(-\sum_{k=1}^p \lambda_k (f_k(x) - f_k(x_n))/T_n)$, draws p randomly and uniformly distributed in $[0, 1]$ and returns $(p < p_n)$.
- **isFinished**, a predicate that checks whether either a predefined number of iterations N_{stop} or a limit temperature T_{stop} in the cooling schedule has been reached.
- **merge**, a function that builds the union $PE \cup PE_\lambda, \lambda \in \Lambda$ eliminating dominated solutions: $\text{argmin}(PE \cup PE_\lambda)$

(2000), a simulation with a fictitious decision maker is reported for the knapsack problem with four objectives and the assignment problem with three objectives. In Ulungu et al. (1998) the interactive version is used for solving a real situation: the problem of homogeneous grouping of nuclear fuel. In Ulungu et al. (1999), different options in the implementation of MOSA are illustrated by the use of extensive experiments. The bi-objective knapsack problem is used as support and discussions are given in comparison with exact solutions. Tuytens et al. (2000) is devoted to the use of MOSA for solving the bi-objective assignment problem. The improvement over the initial solution is discussed. Numerical results are reported and compared with solutions obtained with an exact method. The solution of a variety of scheduling problems with MOSA has been experimented in Loukil Moalla et al. (2000b). Numerical results for bi-criteria single machine, parallel machine, and permutation flowshop problems are reported. Some remarks on the number of potentially efficient solutions are mentioned. Recently a multiple objective Vehicle Routing Problem with time windows has been solved with MOSA, El-Sherbeny (2001). Three categories of objectives are discussed: concerning the vehicles used (number of vehicles, number of covered/uncovered vehicles), concerning time (total duration of the routes, the homogeneity of the duration of the routes, working time not used, total waiting times due to time-window constraints), and concerning the flexible duration of the routes (longer duration of the routes is preferred).

Pareto Simulated Annealing (PSA) by Czyzak and Jaskiewicz (1996), Czyzak and Jaskiewicz (1997), and Czyzak and Jaskiewicz (1998). PSA combines simulated annealing principles with ideas coming from genetic algorithms. The main differences to MOSA concern the management of weights and the consideration of a set of current solutions. A sample $S \subset X$ of $\#S$ solutions is determined and used as initial solutions. Each solution in this set is “optimized” iteratively following the same mechanisms explained above, i.e. by generating neighbouring solutions that may be accepted according to a probabilistic strategy. The authors suggest one of the C (7.1), SL (7.2), or W (7.3) rules for the acceptance probability $P(x, x_n, T_n, \lambda)$. For a given solution $\bar{x} \in S$ the weights are changed in order to increase the probability of moving it away from its closest neighbour in S denoted by $\bar{x}'$. Solutions in S play the role of agents working almost inde-

pendently but exchanging information about their positions in the objective space. Thus, the interaction between solutions guides the generation process through the values of λ . This exploration principle will hopefully lead to an approximation spread uniformly along the non-dominated frontier. On the basis of the results of numerical experiments the way of recomputing the weights has been redefined in later papers by Jaszkiwicz (2001a) and Jaszkiwicz (2001c). An additive formulation has shown best performances and replaces the initial multiplicative form.

The PSA method is published for the first time in 1996 (Czyzak and Jaszkiwicz (1996)), but it has been described in technical reports (from the Institute of Computing Science, Poznan University of Technology) in 1994 and also mentioned in Jaszkiwicz's PhD thesis 1995. Czyzak and Jaszkiwicz (1996) and Czyzak and Jaszkiwicz (1998) are methodological papers where the PSA method is completely described. Numerical experiments on the multiobjective knapsack problem are reported. The method is also extensively described in Jaszkiwicz (2001c). Using the multiobjective knapsack problem, Jaszkiwicz (2001a) presents a comparative experiment of various metaheuristics including the PSA method. The instances have 2, 3, and 4 objectives functions. Hapke et al. (1996), Hapke et al. (1998a) and Jaszkiwicz (1997) describe the use of the PSA method for solving practical problems: project scheduling problems, nurse scheduling, optimization of complex manufacturing systems. In Jaszkiwicz and Ferhat (1999), PSA is incorporated in a method for multiple criteria choice problems. PSA has been coupled with an interactive procedure (light beam search) in order to organize an interactive search in Hapke et al. (1998b). The PSA method has been adapted for solving fuzzy multiobjective combinatorial optimization problems, Hapke et al. (1997), Hapke et al. (2000a) and Hapke et al. (2000b). These papers report applications on fuzzy project scheduling problems with multiple objectives.

Engrand's Method by Engrand (1997), Engrand and Mouney (1998).

The method is introduced during the ICONE conference and revised by Parks and Suppapitnarm (1999), Suppapitnarm and Parks (1999), Suppapitnarm et al. (2000). This method is a hybridization of simulated annealing principles with genetic algorithms. The method has been originally applied to a nuclear fuel management problem.

Engrand (1997) shows that the method does find the trade-off surface and gives performances comparable with a multiobjective GA method on some test cases. Parks and Suppakitnarm (1999) present the revision of Engrand's method (denoted here by MOSA99) and its application to the pressurized water reactor reload core design optimization problem. The main characteristic of this revised version is its ability to work without search directions, using a population of individuals to ensure the exploration of the complete trade-off surface. Each objective is considered separately. The method uses only the non-domination definition to select potentially efficient solutions, thus avoiding the management of search directions and aggregation mechanisms. Advanced strategies use the population of potentially efficient solutions to drive the approximation mechanism, thus ensuring the detection of the whole efficient frontier.

Recommendations are given about the tuning of the parameters in Suppakitnarm et al. (2000). A comparative presentation of results obtained with MOSA99 and the NSGA for an example problem given by Schaffer is reported in Suppakitnarm and Parks (1999). The MOSA99 method is described in detail and refinements are discussed in Suppakitnarm et al. (2000). These concern the solution acceptance rule, annealing schedule, constraints, and a variety of return to base strategies. The method is experimentally verified on three case studies with two and three objectives: a simple problem, a practical problem concerning the deployment system for a rigid panel on a spacecraft, and a problem of optimizing the performance of a ten-bar cantilever truss.

The Trip Planning Problem (Godart (2001)). In his PhD thesis Godart introduces the “Trip Planning Problem”, as a preference-based multiobjective travelling salesman problem with activity and lodging selection. The complete formulation contains five objectives: min transportation cost, min activity cost, min lodging cost, max activity attractivity and max lodging attractivity. A simulated annealing based method, close to the MOSA method, is used to solve this combinatorial problem. Two operational modes are reported: “uni-directional” mode where only one compromise solution is computed and the “omni-directional” mode where several compromise solutions are computed.

Interactive Method for 0-1 Multiobjective Problems by Alves and

Climaco (2000). This is a general interactive method for solving 0-1 multiobjective problems where simulated annealing and tabu search work as two alternative and complementary computing procedures. It is a progressive and selective search of potentially efficient solutions by focusing the search on a sub-region delimited by information specified by the decision-maker for the objective function values. Computational results for multiple-constraint knapsack problems with two objectives are reported.

Bicriteria Scheduling Problems on a Single Machine (Koktener and Köksalan (2000)). An SA algorithm is developed and tested for bicriteria scheduling problems on a single machine using total flow-time with maximum earliness and number of tardy jobs as criteria. Various problem sets (ranges of due date, problem sizes, processing time) are used for the experiment phase. A comparison with previous known results is given.

Other Simulated Annealing-based Methods. Nam and Park (2000) method. This is another simulated annealing-based method. The authors report good results in comparison with MOEA. Applications include aircrew rostering problems, Lučić and Teodorović (1999), assembly line balancing problems with parallel workstations, McMullen and Frazier (1999), and analogue filter tuning, Thompson (2001).

8 The Tabu Search Wave

8.1 Multiobjective Tabu Search by Gandibleux et al. (1996)

In 1996 (MOPGP 96 conference, Torremolinos), Gandibleux et al. introduced a method called MOTS for MultiObjective Tabu Search (Gandibleux et al. (1997)). It is the first TS-based method designed to compute a set of potentially efficient solutions. Using a scalarizing function and a reference point, the method performs a series of tabu processes guided automatically in the objective space by the current approximation of the non-dominated frontier. Intensification, diversification and tabu daemon (usually called aspiration criteria) are designed for the multiobjective case. Two tabu memories are used, one on the decision space $TmemX$, the second on the objective space $TmemY$. The former is an attribute-based tabu list pre-

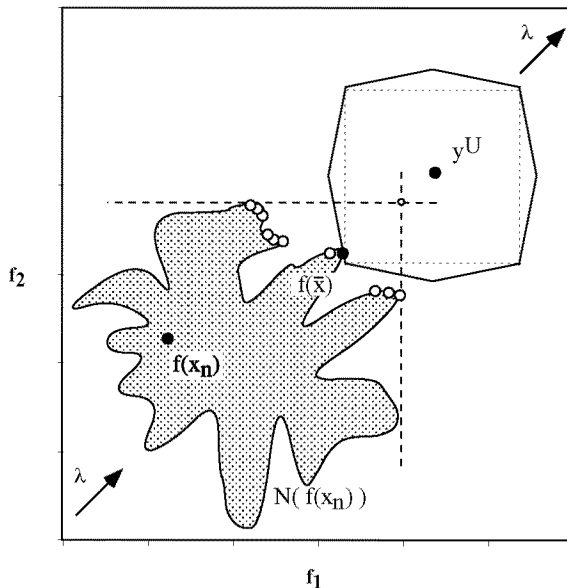


Figure 8: Illustration of the search strategy in MOTS for the bi-objective case. Points $\circ$ are the potentially efficient solutions generated during an iteration.

venting a return to already visited solutions during a tabu process. The latter is connected to the objectives and based on an improvement measure of each objective. It is used for updating weights between two consecutive tabu processes. The method is shown in Algorithm 5.

The before MOTS search strategy is encapsulated in a *tabu process*, which is composed of a series of iterations. Let us consider, at the n^{th} iteration, the current solution x_n and its (sub)neighbourhood $\mathcal{N}(x_n)$ obtained according to a suitable move $x_n \rightarrow x$ defined according to the structure of the feasible domain X (routine `exploreNeighbourhood`). The successor $\bar{x}$ of x_n for the next iteration is selected from the list of neighbour solutions $\mathcal{L} = \{x \in \mathcal{N}(x_n)\}$ as the best according to a weighted scalarizing function $S(f(x), y^U, \lambda)$:

$$S(f(x), y^U, \lambda) = \max_{1 \leq k \leq p} \{ \lambda_k (y_k^U - f_k(x)) \} + \rho \sum_{k=1}^p \lambda_k (y_k^U - f_k(x)), \rho > 0$$

The number of candidates in list $\mathcal{L}$ is limited to K solutions. The value of this parameter depends on the neighbourhood size ($1 \leq K \leq$

size of $\mathcal{N}(x_n)$). The reference point y^U in the scalarizing function is the locally determined utopian point $y^U = (y_1^U, \dots, y_p^U)$ over $\mathcal{L}$, where $y_k^U > \sup\{f_k(x) | x \in \mathcal{L}\}$ (routine `utopianPoint`). This point dominates the ideal point given by the highest objective function value on each objective among the solutions in the neighbourhood of the current solution. The tabu list $TmemX$ is used to avoid cycling (routines `isTabu` and `updateTabuMemoryX`). The selected solution $\bar{x} \in \mathcal{L}$, which minimizes $S(f(x), y^U, \lambda)$ over $\mathcal{L}$ such that the move $x_n \rightarrow \bar{x}$ is not tabu, becomes the new current solution x_{n+1} , as depicted in Figure 8. The *tabu daemon* overrides the tabu status of a solution $x' \in \mathcal{N}(x_n)$, when $s(f(x'), \lambda) \leq s(f(\bar{x}), \lambda) - \Delta$, with Δ being a static or dynamic threshold value (routine `TabuDaemon`). As $\mathcal{L}$ is generally a finite enumerable subset of X , the successor solution x_{n+1} can be found easily. However, the time complexity depends on the size of the neighbourhood $\mathcal{N}(x_n)$. Each iteration ends with the identification of the potentially efficient solutions pe in $\mathcal{L}$, which represents a local approximation of the non-dominated frontier (routine `archive`).

One tabu process is an intensification mechanism in which the effort spent is dynamically adjusted according to pe , the approximation obtained, in comparison with the global approximation PE . The first adjustment is based on a strategy promoting a “promising candidate”. If $\bar{x} \in pe$ seems to be a promising solution, then the current search is increased. The promising characterisation of $\bar{x}$ is evaluated by regarding the number of solutions of PE that were dominated by $\bar{x}$ (routine `isPromising`). The second adjustment is based on the detection of a “sterile search”. When pe contains no potentially efficient solution according to PE (routine `isSterile`), the current neighbourhood is said to be sterile. This means that the neighbourhood is not promising in terms of efficient solutions, and the interest in the current search process is decreased.

The scalarizing function leads to the exploration of the non-dominated frontier in the direction given by the weight vector λ . The diversification strategy updates the weights periodically and automatically so that the importance of objectives that have been significantly improved is decreased (routine `setSearchDirection`). To perform this diversification strategy, a pseudo-criterion preference model is suggested. Using threshold parameters for each objective, which measure the level of improvement $f_k(\bar{x}) - f_k(x)$ according to the objective k , one of the following four scenarios is determined: 1) the performance improvement for objective k is worse (D, dominated); 2) improvement is too weak to be interesting (I, indifferent); 3) improvement

is significant (S, weak improvement) or 4) improvement is strong (P, strong improvement). When the objective k is significantly or strongly improved, the weight λ_k is either slightly or strongly decreased, whereas the weights related to the others cases are increased. To ensure that all the areas of the non-dominated frontier are visited, a tabu list with respect to the objectives is introduced (*TmemY*). In the case S (respectively case P), the objective k is declared tabu for a predetermined number of iterations. Thus, its weight cannot be increased as long as its tabu status is true. Obviously, the diversification scheme is simpler in the case of biobjective problems. A deterministic and myopic design of weights $\lambda \in \Lambda$ can be performed, which provides a suitable description of the search direction.

MOTS is a generic method, rather than a ready-to-use technique. All of its primitives need to be stated in a suitable manner, according to the MOCO problem to be solved.

8.2 Bibliographic Notes

The extension of TS to multiobjective programming is recent in comparison with the other metaheuristics. Hybrid TS-based methods have been proposed, in an effort to improve solution diversification along the non-dominated frontier. Some ideas come from MOEA, like the use of a population (Hansen (2000)), or a combination of tabu search with genetic algorithms (Ben Abdelaziz et al. (1999)). Multiobjective tabu search procedures have been applied mainly to MOCO problems, particularly the knapsack problem. The following MOMH, based on TS, can be found in the literature.

Multiobjective Tabu Search (MOTS) by Gandibleux, Mezdaoui and Fréville, 1996 (Gandibleux et al. (1997)). This method was first instantiated on an unconstrained permutation problem, and then later on the biobjective knapsack problem, Gandibleux and Fréville (2000), in combination with bounds to reduce the search space.

Sun's Method, Sun (1997). An interactive procedure using tabu search for general multiple objective combinatorial optimization problems, the procedure is similar to the Combined Tchebycheff/Aspiration Criterion Vector Method, Steuer et al. (1993). The tabu search is used to solve subproblems in order to find potentially efficient solutions.

Algorithm 5 MOTS, MultiObjective Tabu Search

```

input  :   $TTX$ ,           tabu tenure on decision space
           $TTY, PSI$ ,       tabu tenure and thresholds on
                        objective space
           $\Delta, K$ ,        daemon activation parameter and
                        size of candidate list
           $IterInit$ ,      initial intensity level for a tabu
                        process
           $stopConditions$ , conditions to stop the search
output :   $PE$ ,           the set of potentially efficient
                        solutions

begin MOTS
   $TmemX, TmemY \leftarrow \emptyset$ 
   $n \leftarrow 1$ ; select  $x_n \in X$ ;  $PE \leftarrow \{x_n\}$ 
  --| Diversification in Z
  repeat  $\lambda \leftarrow \text{setSearchDirection}(TmemY, PE)$ 
         $Iter \leftarrow IterInit$ 
    --| Tabu process
    repeat --| Local exploration of the non-dominated frontier
           $\mathcal{L} \leftarrow \text{exploreNeighbourhood}(x_n)$ 
           $\tilde{z} \leftarrow \text{utopianPoint}(\mathcal{L})$ 

          --| Selection of the successor solution
           $\bar{x} \leftarrow \arg \min \{s(f(x), y^U, \lambda) \mid x \in \mathcal{L},$ 
                                 $\neg \text{isTabu}(\text{move}(x_n \rightarrow x), TmemX)\}$ 
           $x^* \leftarrow \text{TabuDaemon}(\bar{x}, \mathcal{L}, TmemX)$ 
          if  $x^* = \emptyset$  then  $x_{n++} \leftarrow \bar{x}$  else  $x_{n++} \leftarrow x^*$  endif
           $TmemX \leftarrow$ 
             $\text{updateTabuMemoryX}(\text{move}(x_{n-1} \rightarrow x_n), TmemX)$ 

          --| Dynamic adjustment of the intensification
          --| and update of the approximation
           $pe \leftarrow \{x_n\}; \forall x \in \mathcal{L} : pe \leftarrow \text{archive}(pe, x)$ 
          if  $\text{isPromising}(x_n, PE)$  then increase  $Iter$  endif
          if  $\text{isSterile}(pe, PE)$  then decrease  $Iter$  endif
           $PE \leftarrow \text{merge}(PE, pe); Iter \leftarrow Iter - 1$ 

    until  $Iter = 0$  --| iterations are performed

  until  $stopConditions$  are fulfilled
end MOTS

```

The principles used for designing the TS search strategy are similar to those defined for MOTS. This method has been used for facility location planning, Agrell et al. (1999).

Multiobjective Tabu Search (MOTS97) by Hansen (2000). This method uses a set of “generation solutions”, each with its own tabu list. These solutions are dispersed throughout the objective space in order to allow searches in different areas of the non-dominated frontier. Weights are defined for each solution with the aim of forcing the search into a certain direction of the non-dominated frontier and away from other current solutions that are efficient with respect to it. Diversification is ensured by a set of generation solutions and a drift criterion. Results for the knapsack problem are available, and also for the resource constrained project scheduling problem in Viana and Sousa (2000).

Ben Abdelaziz, Chaouachi and Krichen’s Hybrid Method, Ben Abdelaziz et al. (1999). The authors present a multiobjective hybrid heuristic for the knapsack problem. The method is a mix of a tabu search and a genetic algorithm.

Baykasoglu, Owen and Gindy’s Method, Baykasoglu et al. (1999). A candidate list provides an opportunity to diversify the search. The method is designed to handle any type of variable (integer, zero-one, continuous and mixed), and has been used for goal programming problems, Baykasoglu (2001a) and Baykasoglu (2001b).

Other tabu search-based methods have been developed for scheduling problems (Loukil Moalla et al. (2000a)) and the trip planning problem, Godart (2001). A hybrid and interactive solution process based on SA and TS is proposed in Alves and Climaco (2000).

9 The Ant Colony Optimization Wave

Recently the Ant Colony System (ACS) paradigm, a population based algorithm, has been adapted to multiobjective optimization. Ant colony optimization techniques are inspired by the behavior of real ants foraging for food. The key to the effectiveness of a colony of ants in finding short

paths to a food source is a chemical substance called pheromone. It provides the ants with the ability to communicate. Ants will initially move along random directions depositing pheromone along their paths. When an ant finds a food source it returns to the nest. An ant on a shortest path will return to the nest first, thus more pheromone will be deposited on the shortest paths. Because a moving ant will choose a path with a probability that depends on the amount of pheromone detected, paths that are more frequently travelled become more attractive, and over time the shortest paths will be used most often. Also, the pheromone evaporates over time, so that pheromone trails of infrequently travelled paths become weaker and those of frequently travelled shortest paths are reinforced.

The use of artificial ant colony systems has first been proposed by Dorigo and co-authors. Details of single objective optimisation techniques based on ant colonies can be found, e.g. in Dorigo (1992), Dorigo et al. (1996), and Dorigo et al. (1997). In recent years (since 1999) ant colony optimization technique have also been proposed for multiobjective combinatorial optimization. Because of the way ant colony systems work, their application is most attractive for problems for which solutions can be constructed sequentially, and all applications in the MOCO area to date are of such a type: TSP, VRP, sequencing and scheduling problems, and portfolio selection problems.

9.1 Multiobjective Ant Colony Optimization Algorithms

Multiobjective ant colony optimization algorithms have been proposed for problems with a weighted sum of objectives, Doerner et al. (2001b), a hierarchy on the objectives (lexicographic optimization, Gambardella et al. (1999), Gravel et al. (2002), T'Kindt et al. (2002)) and in the Pareto optimality sense, Doerner et al. (2004), Iredi et al. (2001), Mariano and Morales (1999b), McMullen (2001), and Shelokar et al. (2000). They either use a single colony (Doerner et al. (2004), Gravel et al. (2002), Iredi et al. (2001), McMullen (2001), Shelokar et al. (2000), T'Kindt et al. (2002)) or multiple colonies (Doerner et al. (2001b), Gambardella et al. (1999), Iredi et al. (2001), Mariano and Morales (1999b)). Most of the algorithm use both heuristic information, which is problem specific, and pheromone information to calculate the probability for making the next step in the solution construction procedure. As a typical example of a multicolony system for finding efficient solutions we present an algorithm adapted from

Iredi et al. (2001), see Algorithm 6.

Algorithm 6 Multicolony Ant Algorithm by Iredi et al. (2001)

input : m, a , the size of all colonies and one colony
 α, β , weight of pheromone and heuristic information
 output : PE , the set of potentially efficient solutions

begin MCAA

```

 $PE \leftarrow \emptyset$ 
repeat for colony  $l = 1, \dots, \frac{m}{a}$  loop
    initialisePheromone( $\tau$ )
    initialiseHeuristic( $\eta$ )

    for ant  $k = 1, \dots, a$  loop
        assignWeight( $k, l, \lambda$ )
        constructSolution( $k, l, p(\lambda, \alpha, \beta), \tau, \eta, x_k^l$ )
        localPheromoneUpdate( $\tau, x_k^l$ )
    end for

    end for
     $PE \leftarrow \text{nonDominated}(PE, \cup_{k,l} \{x_k^l\})$ 
    globalPheromoneUpdate( $\tau, \cup_{k,l} \{x_k^l\}, PE$ )

```

until stoppingCriterion

end MCAA

The details of the algorithm are problem dependent. In the scheduling problem discussed in Iredi et al. (2001), pheromone and heuristic information is stored in square matrices, where the indices of η_{ij} and τ_{ij} refer to having job j in position i in the sequence. Heuristic information is computed according to the objectives and remains unchanged during the algorithm. Initially, $\tau_{ij} = \tau_0$. Let m be the total number of ants and a be the number of ants per colony.

Several methods for the choice of weights λ and k of colony l uses for the first (of two) objectives are proposed.

$$\begin{aligned}\text{assignweight}(k, l, \lambda) &= \frac{k-1}{(m/a)-1} \\ \text{assignweight}(k, l, \lambda) &= \frac{l-1}{a} + \frac{k}{m} \\ \text{assignweight}(k, l, \lambda) &= \frac{i-1}{a+1} + \frac{2(k-1)}{(a+1)((m/a)-1)}.\end{aligned}$$

In the first case, the ants in each colony use the same weights, in the second the weight intervals for the colonies have only the boundaries in common, whereas in the last the weight interval of colony l intersects with those of colony $l-1$ and $l+1$.

The probability by which job j is chosen to be in position i is given by

$$p_{ij} = \frac{\tau_{ij}^{1\lambda\alpha} \tau_{ij}^{2(1-\lambda)\alpha} \eta_{ij}^{1\lambda\beta} \eta_{ij}^{2(1-\lambda)\beta}}{\sum_{h \in S} \tau_{ih}^{1\lambda\alpha} \tau_{ih}^{2(1-\lambda)\alpha} \eta_{ih}^{1\lambda\beta} \eta_{ih}^{2(1-\lambda)\beta}},$$

where α and β are weights of pheromone and heuristic information, respectively.

All ants that generated a solution in PE in the current iteration contribute to the pheromone update. First τ_{ij} is set to $(1-\rho)\tau_{ij}$ for all (i, j) (evaporation). Then $\tau_{ij}^h \leftarrow (1-\rho)\tau_{ij}^h + 1/r$, for all (i, j) appearing in the r updating ants. Since each colony maintains its own pheromone matrices, the update can either be done by colony or by region in PE . No local pheromone update is done.

The algorithm of Doerner et al. (2004) for portfolio selection also fits in this framework. Here $p = 1$, pheromone is initialised to $\tau_0 > 0$, heuristic information $\eta_i(x)$ is based on feasibility, depends on the partial solution constructed so far, and is updated during the algorithm. The authors use a random assignment of weights to cope with the many objectives.

Local pheromone update is $\tau_i^k \leftarrow (1-\rho)\tau_i^k + \rho\tau_0$ for objective k and project i , if project i has been added to a portfolio by an ant. The ants who contributed a solution to PE in the current iteration update the pheromone globally using $\tau_i^k \leftarrow (1-\rho)\tau_i^k + \rho\Delta_i^k$, where Δ_i^k is 10, if project i appears in the best solution for objective k and 5 if it is in the second best solution.

The solution is constructed by adding project i to the portfolio if a uniformly distributed random number q is not bigger than a parameter q_0 and $\sum_{k=1}^p (\lambda_k \tau_i^k)^\alpha (\eta_i(x))^\beta$ is maximal among all projects that can be added to the partial solution x and have $\eta_i(x) > 0$. Otherwise the project to be added is chosen randomly with probability

$$p_i = \frac{\sum_{k=1}^p (\lambda_k \tau_i^k)^\alpha \eta_i(x)^\beta}{\sum_{h \in S(x)} \sum_{k=1}^p (\lambda_k \tau_h^k)^\alpha \eta_h(x)^\beta}.$$

9.2 Bibliographic Notes

Shelokar et al. (2000), Shelokar et al. (2002), Shelokar et al. (2003): In Shelokar et al. (2000) an ant algorithm for multiobjective continuous optimization is proposed. An interesting feature is that it combines the ant system methodology with the strength Pareto fitness assignment of Zitzler and Thiele (1998) and clustering methods. The algorithm is applied to reliability engineering problems in Shelokar et al. (2002) and to the optimization of reactor regenerator systems in Shelokar et al. (2003).

McMullen (2001): In this paper a just-in-time sequencing problem with the objective of minimizing the number of setups and minimizing the usage rates of raw materials is addressed. The problem is reformulated as a TSP by spatialising the data and applying a standard single objective ant colony algorithm.

Gambardella et al. (1999): Here, a bicriteria vehicle routing problem with time windows with the lexicographically sorted objectives of minimizing the number of vehicles and minimizing total travel distance is solved by a multicolony ant system called MACS. The algorithm uses one colony for each objective, local and global pheromone update. The colonies cooperate through the use of the global best solution for the global pheromone update. Local search is applied to improve the quality of each solution found. Numerical tests show that the algorithm improves some of the best known solutions of test problems from the literature.

Mariano and Morales (1999a), Mariano and Morales (1999b): The algorithm MOAQ uses one ant colony for each objective function. All colonies have the same number of ants. (Partial) solutions of

each colony are used in the next colony. The algorithm is applied to two literature problems (and compared with VEGA (Schaffer (1984))). The research is motivated by the real world problem of designing a water distribution network for irrigation to minimise network cost and maximise profit.

Iredi et al. (2001): The authors propose a number of ant colony optimization algorithms to solve bicriteria combinatorial problems, including ones with single and multiple colonies. Various methods for pheromone update and weight assignment (in order to browse the whole non-dominated frontier) are proposed and tested on a single machine scheduling problem to minimise total tardiness and changeover cost. Numerical tests are presented.

Doerner et al. (2001a), Doerner et al. (2002), Doerner et al. (2003), Doerner et al. (2004): The problem framework is that of a multiobjective portfolio selection problem. The authors consider a rather large number of objectives $p = (B + R)T$, where B is the number of benefit categories, R is the number of resources, and T is the planning period. They use one colony and a random selection of weights of objectives for each ant. The global update considers the best and second best solutions for each objective found in the current iteration. The results on test problems are compared with NSGA (Srinivas and Deb (1994)), PSA (Czyzak and Jaszkievicz (1996)), and the true efficient set (for small problems).

Doerner et al. (2001b): The authors solve a special case of the pickup and delivery problem with the linearly combined objectives of total number of vehicles and empty vehicle movements by a multicolony approach. The colonies use different heuristic information, and their sizes change during the algorithm.

T'Kindt et al. (2002): A (single) ant colony optimization approach is proposed for a two machine bicriteria flowshop problem to minimize makespan and total flowtime in a lexicographic sense. The solutions produced by the ants are improved by local search.

Gravel et al. (2002): The problem of sequencing orders for the casting of aluminium provides the background in this paper. Four objectives are considered in a lexicographic sense. A distance function based on

penalties for bad performance is used to translate the problem into a TSP setting. Global pheromone update considers only the primary objective.

10 Other Approaches and the Trend

In addition to the previously discussed multiobjective versions of the now classic metaheuristics (GA, SA, and TS), and the now very popular ant colony systems, there are other MOMH. Several of them have been published most recently or have until now been discussed only during international conferences:

- The first works using Artificial Neural Networks (ANN) to solve MOP have been published (Malakooti et al. (1990), Sun et al. (1996), Sun et al. (2000)) at the beginning of the Nineties. However, the ANN approach remains marginal.
- Adaptations of metaheuristics, such as the Greedy Randomized Adaptive Search Procedure GRASP, Gandibleux et al. (1998), and Scatter Search, Beausoleil (2001), have been presented at recent international conferences.
- Papers concerning a dedicated heuristic, Köksalan (1999), a stochastic search method (Sysoev and Dolgui (1999)) and a comparison of neighbourhood search techniques for MOP, Marett and Wright (1996), are also worthy of mention.

Combinatorial optimisation is rich in term of results. It is natural to bridge between the known theoretical results for single objective combinatorial problems and MOCO. Managed inside a MOMH, this information can be used advantageously to enforce the aptitude of approximation methods. For example, adding an additional constraint to the unidimensional biobjective knapsack problem will help to reduce the search space, Gandibleux and Fréville (2000).

Another promising way is to design MOMH components that take the specific aspect of MOCO into account. In Gandibleux et al. (2004), similarities in solutions and subsets of exact solutions are used advantageously

by the components of an evolutionary method. Here, interesting performance results are measured with a *path relinking* operator, Glover and Laguna (1997), given a subset of optimal solutions (or approximations) in the initial population. Path-relinking generates new solutions by exploring the trajectories that connect good solutions. A path-relinking operation starts by randomly selecting I_A (the initiating solution) and I_B (the guiding solution), two individuals from the current population (Figure 9). The path-relinking operation generates a path $I_A(=I_0), I_1, \dots, I_B$ through the neighbourhood space, such that the distance between I_i and I_B decreases monotonically in i , where the distance is defined as the number of positions for which different values are assigned in I_i and I_B .

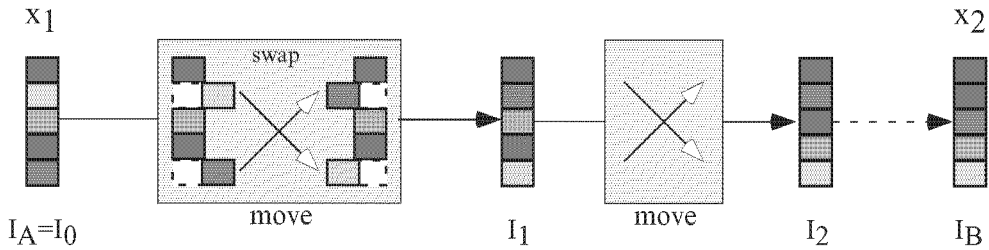


Figure 9: Path-relinking operator principle.

Although many such paths may possibly exist, one path is chosen using, for example, random moves based on a swap operator. Such randomness introduces a form of diversity to the solutions generated along the path. For every intermediate solution I_i , a single solution is generated in the neighbourhood (Figure 10). This principle has been successfully implemented for computing the complete non-dominated frontier of assignment problems with two objectives. Figure 11 provides a sample of these results. Using the same numerical instances, this population-based method based on specific operators outperforms MOSA (Tuytens et al. (2000)).

MOCO problems rarely tackled hitherto are now investigated with MOMH. These include timetabling problems (Paquete and Fonseca (2001)), space allocation problems (Pires et al. (2001)), multi-period distribution management problems (Ribeiro and Lourenço (2001)), and vehicle routing problems (Geiger (2001) and Rahoual et al. (2001)), to name a few.

The last aspect of research concerns issues related to computer implementation. Efficient data structures, such as the quad-tree, have proven

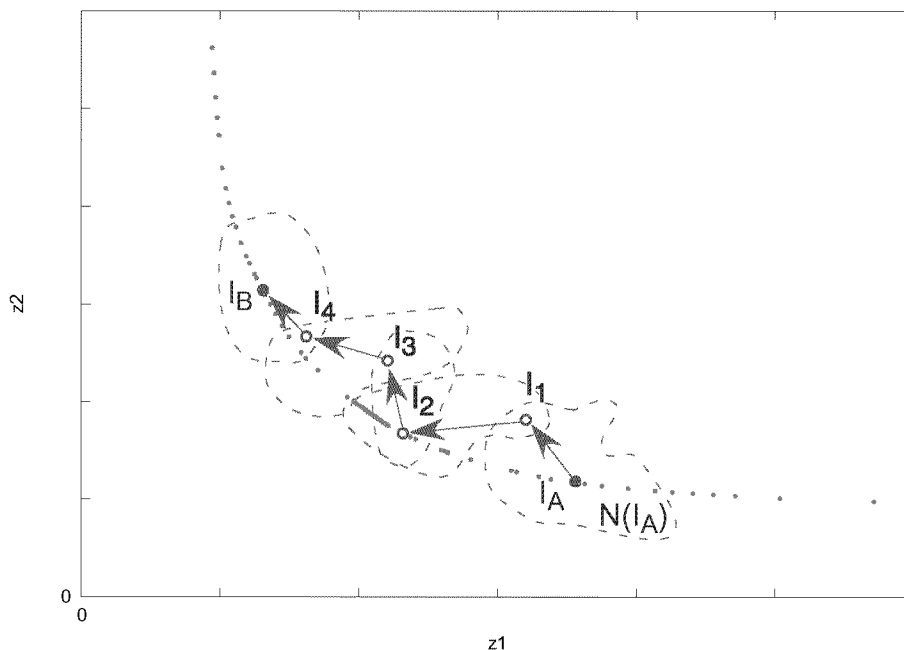


Figure 10: Illustration of a possible path construction. I_A and I_B are two individuals randomly selected from the current population (small bullets). I_A is the initiating solution, I_B is the guiding solution. $N(I_A)$ is the feasible neighbourhood according to the move defined. $I_A - I_1 - I_2 - I_3 - I_4 - I_B$ is the path that is built.

their efficiency for managing non-dominated criterion vectors, Habenicht (1983) and Sun and Steuer (1996). Reusable software, such as object-oriented frameworks for multiobjective local search, is under development, Claro and Sousa (2001).

These new trends promise many future papers.

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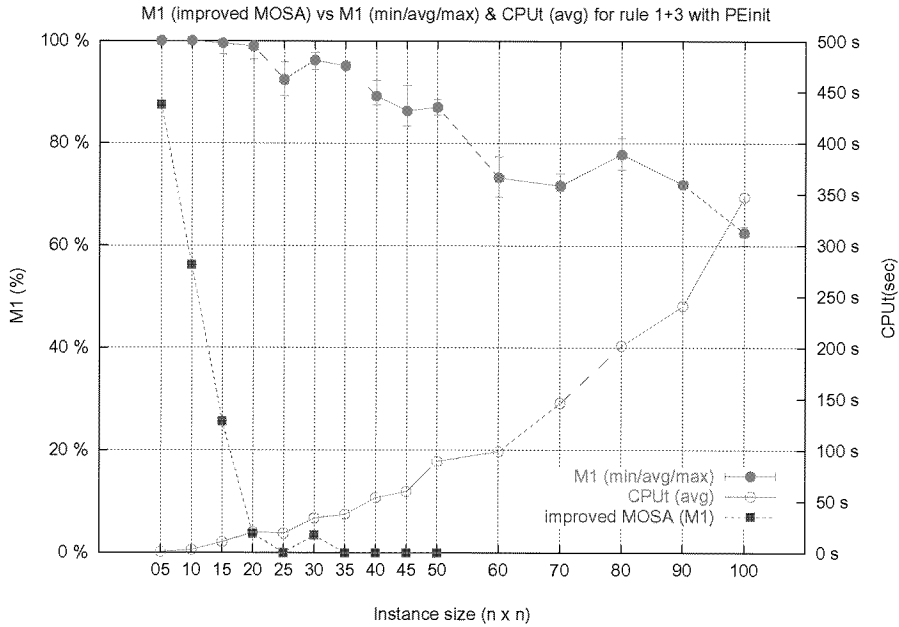


Figure 11: Approximations obtained with the MOSA method and a population-based method for the assignment problem with two objectives. The comparison is based on the performance measure M_1 , Ulungu (1993), which measures the percentage of exact solutions included in the final approximation set.

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